Combined Literature Review - AI and Machine Learning in the Emergency Department (ED) Triage

Artificial intelligence (AI) and machine learning (ML) are revolutionising emergency departments by transforming triage from a static and subjective assessment into a dynamic and data-driven process. The application of AI to the ED has attracted growing research interest, which is driven by well documented limitations of conventional triage systems. These include, undertriage, overtriage and inconsistencies across reviewers as well as the operational pressures of ED overcrowding.

Yi, Baik, and Baek (2024) conducted a systematic review to assess the effects of AI-based triage in the clinical ED setting. In 2023, seven databases were searched with three independent researchers assessing records using the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) protocol, resulting in seven prospective studies from an initial 1,633 records. Most studies used Machine Learning (ML), which enables computers to learn patterns from data, while on study used fuzzy logic, a method that supports decision-making when information is uncertain. These models achieved triage prediction accuracies ranging from 80.5% to 99.1%. Yi, Baik, and Baek (2024) concluded that AI shows promise as a decision-support tool for triage nurses, but the limited evidence and need for further validations means it is not ready for widespread clinical use.

Porto (2024) builds on this by reviewing which ML and Natural Language Processing (NLP) algorithms perform best when assigning patients to triage categories. NLP enables computers to analyse and interpret written language and was examined because of its potential in improving systems such as the Manchester Triage Scale (MTS). Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, 60 studies were done which covered 57 ML algorithms. Logistic Regression was the common model used, while XGBoost, Gradient Boosting, and Deep Neural Networks, achieved the strongest performance. XGBoost is a prediction-focused ML algorithm, Gradient Boosting improves accuracy by combining multiple decision trees, and Deep Neural Networks learn complex matters from large datasets. The most predictive variables were oxygen saturation, chief complaints, systolic blood pressure, age and the mode of arrival. While the NLP methods improved triage classifications when combined with standard patient data, there is a high risk of bias and significant differences between studies which limit the strength of the outcomes.

Patipan Sitthiprawiat et al. (2025) moves from reviewing existing evidence to developing and testing other models. The study analysed 163,452 ED visits from a hospital in Thailand between 2018-2022. XGBoost, Logistic Regression and Random Forest models were compared with the Canadian Triage and Acuity Scale (CTAS). Chief complaints were analysed using multilingual sentence embeddings which is done by using an NLP technique that converts written text to a numerical format that the AI can process. The results found that XGBoost outperformed CTAS, by achieving an Area Under the Receiver Operating Characteristic (AUROC), a measure of how well a model distinguishes between outcomes, of 0.917 compared with 0.882 and an Area Under the Precision-Recall Curve (AUPRC), a measure particularly useful for rare outcomes, of 0.629 compared 0.333. The strongest predictors identified were mode of arrival, age and chief complaints. The main limitation according to (Patipan Sitthiprawiat et al. 2025) is that the model only considered patients admitted directly to the ICU, excluding those that were transferred from general wards. This may have overlooked triage patients and lead to an overestimation of the models performance.

Across all three papers, several common themes emerge. Chief complaints, age and vital signs were consistently identified as important predictors in triage outcomes, while AI-based approaches frequently outperformed traditional triage systems. However, the studies also highlighted important limitations, including small evidence bases, potential bias and limited validation across different clinical settings. These findings suggest that AI-assisted triage has considerable potential, however, further evaluation is required before widespread implementation can be recommended.

References

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